Manuscript Intelligence Archive — Website Blueprint

Tech stack: PHP (8.x) · MySQL (8.x) · HTML5 · CSS3 · Vanilla JavaScript

Status: Sirf blueprint (build baad me)

1. Project Overview

Ek digital platform jahan purane manuscripts upload hote hain, unka text nikalta hai, modern bhasha me translate hota hai, events ki timeline banti hai, places map par dikhte hain, aur alag-alag manuscripts ke beech chhupe hue connections automatically identify hote hain.

Target users: Researchers, historians, students, librarians, museum/archive staff, general public.

2. Core Features (aapne bataye)

2.1 Manuscripts se Events ki Timeline

- Har manuscript ke text se events (naam, tithi/saal, jagah, log) extract hote hain.
- Pehle **manual + semi-automatic** entry (researcher events add/approve karta hai); baad me auto-suggestion.
- Date handling: exact date, approximate ("c. 1200 CE"), range, aur alag calendars (Vikram Samvat, Shaka, Hijri, Gregorian) — sab ko ek normalized `sort_year` me convert karke store karna.
- Frontend: zoomable horizontal timeline, filter by manuscript / person / place / category (war, treaty, trade, religion...).
- Library: **vis-timeline** ya custom JS + CSS.

2.2 Ancient Text ka Modern Language me Translation

- Original text (Sanskrit, Persian, Prakrit, Pali, Old Hindi, Arabic, Latin etc.) → Modern Hindi / English.
- Workflow: transcription → original text → machine translation (draft) → human reviewer correction → "Verified" tag.
- **Side-by-side view:** original | transliteration | modern translation, line-by-line sync.
- Har translation ka version history (kisne, kab, kya badla).
- Translation engine: PHP se external AI/translation API call (cURL), API key `.env` me. Result DB me cache hota hai taaki baar-baar cost na lage.
- Glossary: purane shabdon ka arth (word → meaning table), hover par tooltip.

2.3 Historical Places ko Map par Visualize karna

- Har place ka latitude/longitude, aur **ancient name vs modern name** (e.g., Pataliputra → Patna).
- Interactive map: **Leaflet.js + OpenStreetMap** (free, no billing).
- Features: marker clusters, manuscript/event ke hisaab se filter, time-slider (us saal me kaun si jagah relevant thi), route/path lines (yatra, sena ka marching route, trade route).
- Place par click → us jagah se jude manuscripts, events, log.

2.4 Rare Documents ka Searchable Digital Archive

- Upload: PDF / JPG / PNG / TIFF scans + metadata.
- Metadata: title, author, language, script, era, origin place, repository/source, condition, rights/license, call number.
- **Search:** MySQL FULLTEXT (original text + translation + metadata), plus filters (language, era, place, person, type). Typo tolerance ke liye LIKE/soundex fallback.
- Viewer: zoom/pan image viewer (OpenSeadragon), page-by-page navigation.
- Access levels: public / restricted / private.

2.5 Different Manuscripts ke Connections se Unnoticed Historical Links

Ye platform ka sabse unique feature hai.

Approach (step-by-step, PHP + MySQL friendly):

1. **Entity extraction:** har manuscript se persons, places, organizations, dates, objects tag hote hain (`entities` table).
2. **Shared-entity linking:** agar do alag manuscripts me same person/place/event aata hai → candidate link.
3. **Text similarity:** TF-IDF / keyword overlap score; similar phrases, similar names ke variants (spelling variations — Soundex/Levenshtein).
4. **Time + place proximity:** same period + same region wale documents ko score boost.
5. **Link score (0–100)** ke saath "kyun connected hai" ka explanation (shared entities list).
6. **Review queue:** researcher link ko Accept / Reject / Comment karta hai. Accepted links "Confirmed" ho jaate hain.
7. **Graph visualization:** manuscripts/persons/places ek network graph me (**vis-network** ya D3.js) — cluster, central nodes, isolated nodes dikhte hain.
8. **"Hidden Link" alerts:** naye upload ke baad system suggest karta hai: "*Ye manuscript X aur Y se judta lagta hai.*"

- Optional upgrade: AI (LLM) se semantic comparison — alag script/language me likhe documents ke beech meaning-based match.

3. Extra Features (sirf 5 important aur heavy)

3.1 OCR / HTR Pipeline + Transcription Workspace

- Scan upload hote hi background job me OCR/HTR chalta hai (Tesseract via PHP exec, ya external OCR API), taaki manual typing bache.
- Transcription workspace: image ek taraf, editable text dusri taraf; line-by-line correction, uncertain words highlight.
- Har page ka status: *OCR draft → Corrected → Verified*, confidence score ke saath.
- Ye pipeline translation, search aur link discovery — teeno ka base hai.

3.2 Persons & Genealogy

- Har vyakti (raja, lekhak, vyapari, sant) ka profile: alternate names, jeevan kaal, role, bio, jude manuscripts/events/places.
- Relationships: parent-child, guru-shishya, ally, rival, patron — ek family/relationship tree ke roop me.
- Persons timeline aur map se jude hote hain (kaun, kab, kahan tha).

3.3 Annotation Tools + Scholarly Discussion

- Manuscript image par region highlight, aur text par line/word selection ke saath note.
- Har annotation par threaded comments (researchers ki behas), source/citation attach karna.
- Annotation types: *Question, Correction, Reference, Hypothesis.*
- Public / private annotation visibility.

3.4 Entity Resolution (Fuzzy Name Matching + Merge)

- Ek hi vyakti/jagah ke alag spellings aur scripts (Akbar / Akbarr / □□□□ / اکبر) automatically detect karna.
- Matching: Soundex, Levenshtein, transliteration tables, context (same era/place).
- Admin/editor ke liye duplicate-merge screen: suggested matches, side-by-side compare, merge/undo.
- Isse link discovery (2.5) ki quality directly badhti hai.

3.5 Public REST API + Export

- Read-only API: manuscripts, events, places, persons, links — JSON format, API key + rate limit ke saath.
- Export: translation/transcription PDF, events CSV/JSON, citation (APA/MLA/Chicago), timeline aur map ki image.
- Dusre researchers, apps aur museums apna kaam is data par bana sakte hain.

4. User Roles & Permissions

Role	Permissions
Visitor	Public archive search, read, map/timeline dekhna
Registered User	Annotations aur comments, API key lena
Researcher/Editor	Transcription/translation verify, events approve, links confirm, entity merge
Admin	Users, uploads, settings, full control

5. System Architecture

```
Browser (HTML/CSS/JS)
   │  fetch() / AJAX (JSON)
   ▼
PHP Backend (MVC-style, REST endpoints)
   ├── Auth & Roles
   ├── Upload & File Handler
   ├── OCR Service (Tesseract / API)
   ├── Translation Service (API + cache)
   ├── Entity Extraction + Resolution Service
   ├── Link Discovery Engine (scoring)
   └── Search Service (FULLTEXT)
   ▼
MySQL Database  +  /storage (scans, derived images)
```

Background jobs: OCR, translation, entity matching, link-discovery heavy kaam hain → `jobs` table + cron (CLI PHP script) se queue process karna, taaki website slow na ho.

6. Folder Structure

```
/manuscript-archive
├── /public                # web root
│   ├── index.php
│   ├── /assets
│   │   ├── /css
│   │   ├── /js
│   │   └── /img
│   └── /api               # JSON endpoints entry
├── /app
│   ├── /controllers
│   ├── /models
│   ├── /services          # ocr, translate, linker, resolver, search
│   ├── /views             # PHP templates
│   └── /helpers
├── /config                # db.php, app.php
├── /storage
│   ├── /originals         # raw scans
│   └── /derivatives       # thumbnails, tiles
├── /cron                  # queue worker, link scan
├── /database              # schema.sql, seed.sql, migrations
├── /docs
└── .env
```

7. Database Design (MySQL)

Table	Key Columns
`users`	id, name, email, password_hash, role, api_key, created_at
`manuscripts`	id, title, author, language_id, script, era_start, era_end, origin_place_id, source_repository, call_number, license, access_level, status, uploaded_by, created_at
`manuscript_pages`	id, manuscript_id, page_no, image_path, ocr_text, ocr_confidence, transcription, status
`languages`	id, name, script, iso_code
`translations`	id, page_id, target_lang, text, method (AI/human), status (draft/verified), reviewer_id, version, created_at
`glossary`	id, term, language_id, meaning_hi, meaning_en, notes
`persons`	id, name, alt_names, birth_year, death_year, role, bio
`person_relations`	id, person_a, person_b, relation_type, source_manuscript_id
`places`	id, ancient_name, modern_name, lat, lng, region, country, notes
`events`	id, title, description, date_text, sort_year, end_year, calendar_type, place_id, category, confidence, status
`event_sources`	event_id, manuscript_id, page_id, quote_ref
`entities`	id, type (person/place/org/object), name, normalized_name, ref_id, merged_into
`entity_mentions`	id, entity_id, manuscript_id, page_id, position, context_snippet
`links`	id, manuscript_a, manuscript_b, score, reason_json, status (suggested/confirmed/rejected), reviewed_by, created_at
`annotations`	id, page_id, user_id, type, coordinates/range, note, visibility, parent_id, created_at
`jobs`	id, type, payload, status, attempts, created_at

Indexes: FULLTEXT on `manuscripts(title, author)`, `manuscript_pages(ocr_text, transcription)`, `translations(text)`; B-tree on `events.sort_year`, `places(lat,lng)`, `links.score`.

8. Pages / Screens

Public: Home · Search results · Manuscript detail (viewer + translation + annotations) · Timeline · Map · Connections graph · Person page (with relationship tree) · Place page · About / API docs

User: Login/Register · Profile · My annotations

Researcher/Admin: Upload manuscript · Transcription editor · Translation review queue · Event editor · Link review queue · Entity merge screen · User management · Settings

9. API Endpoints (JSON, indicative)

```
GET  /api/manuscripts?q=&lang=&era=&place=
GET  /api/manuscripts/{id}
POST /api/manuscripts                 (upload)
GET  /api/manuscripts/{id}/pages
PUT  /api/pages/{id}/transcription
POST /api/pages/{id}/translate
PUT  /api/translations/{id}/verify
GET  /api/events?from=&to=&place=&category=
POST /api/events
GET  /api/places?bbox=
GET  /api/persons/{id}
GET  /api/persons/{id}/relations
POST /api/pages/{id}/annotations
GET  /api/links?manuscript={id}&status=
PUT  /api/links/{id}/review
GET  /api/entities/duplicates
POST /api/entities/merge
GET  /api/graph?root={id}&depth=2
GET  /api/search?q=
GET  /api/export/{type}/{id}
```

10. Frontend Libraries (free)

- **Leaflet.js** — maps
- **vis-timeline** — timeline
- **vis-network / D3.js** — connection graph aur relationship tree
- **OpenSeadragon** — manuscript image zoom viewer
- Pure CSS (Flexbox/Grid), responsive; Hindi fonts (Noto Sans Devanagari)

11. Security & Quality

- Password hashing (`password_hash`), CSRF tokens, prepared statements (PDO) — SQL injection se bachav
- File upload validation (type, size, MIME check), uploads web-root ke bahar
- Output escaping (XSS), rate limiting on search/API
- HTTPS, secure sessions, `.env` me secrets
- Role checks har endpoint par
- Copyright/licence metadata aur takedown process

12. Development Roadmap (jab build karna ho)

Phase	Kaam	Time (approx.)
1	DB schema, auth, roles, upload, manuscript CRUD	2–3 hafte
2	Archive viewer + FULLTEXT search + filters	2 hafte
3	OCR pipeline + transcription workspace + translation workflow	3–4 hafte
4	Places + Map	1–2 hafte
5	Events + Timeline (map sync ke saath)	2 hafte
6	Persons & Genealogy + Entity resolution	3 hafte
7	Link discovery + Graph	3–4 hafte
8	Annotations + Public API + Export	2–3 hafte
9	Testing, security audit, performance, deployment	2 hafte

MVP (pehla release): Phase 1 + 2 + 3 + basic Map & Timeline.

13. Risks & Notes

- **OCR accuracy** purani scripts par kam ho sakti hai → human correction zaruri.
- **AI translation** me galtiyan ho sakti hain → hamesha "Draft / Verified" label.
- **Link discovery** sirf *suggestion* deta hai, final faisla researcher karta hai.
- PHP shared hosting par heavy jobs slow honge → VPS recommend.
- Rare documents ke copyright/ownership rules pehle clear karein.